

序列比对

1. 全局比对: $F(i, j)$: $x_1 \dots i$ 与 $y_1 \dots j$ 间的最优匹配值.

$$F(i, j) = \max \begin{cases} F(i-1, j-1) + s(x_i, y_j), \\ F(i, j-1) + d, \\ F(i-1, j) + d \end{cases} \quad \text{Needleman - Wunsch}$$

$F(0, 0) = 0$; 从最大值回溯到 $(0, 0)$

2. 局部比对.

$$F(i, j) = \max \begin{cases} F(i-1, j-1) + s(x_i, y_j) \\ F(i, j-1) + d \\ F(i-1, j) + d \\ 0 \rightarrow \text{若 } F < 0 \text{ 则重新开始比对.} \end{cases} \quad \text{Smith - Waterman}$$

$F(0, 0) = 0$; 从最大值回溯到首个 0

3. 半全局比对 (e.g., 子序列比对)

边缘空位不扣分

$$F(i, j) = \max \begin{cases} F(i-1, j-1) + s(x_i, y_j), \\ F(i, j-1) + d, \\ F(i-1, j) + d \end{cases}$$

$F(i, 0) = 0, F(0, j) = 0$.

从最大值回溯到 $(0, 0)$

复杂度分析:

k 个序列长为 n 的序列 $\Rightarrow k$ 维 dp 矩阵, 状态数 $O(n^k)$

转移至每个状态的路径数 $O(2^k - 1) = O(2^k)$

每条路径上的计算 $O(k^2)$?

$\Rightarrow O(n^k \cdot 2^k \cdot k^2)$

4. 多序列比对 CLUSTAL W: bp + 层次聚类

① Distance matrix

利用双序列全局比对得到任意两序列间的“距离” (e.g., -dp 值)

② Guide tree UPGMA

贪心: 每次归并两个距离最近的, 更新距离,

重复直至归并完成.

③ MSA

基于 guide tree 给出的序列间距离信息渐进比对

使用 PSSM 对得到的 MSA 打分

$\text{score}_i = \log \text{odds ratio} = \log P(x_i | M) - \log P(x_i | R) \Rightarrow \sum_i \text{score}_i$ (全序列总和或最大连续部分和).

$$P(x | M) \approx \hat{\text{Freq}} = \frac{N_{\text{obs}} + 1}{\sum N_{\text{obs}} + C}, \quad C = \# \text{ of classes.}$$

$\downarrow$
pseudocount $\Rightarrow$ 避免 $\log 0$.

$$P(x | R) = \frac{1}{C}.$$

基于贪心的启发式算法, 不保证最优.

序列注释

1. MM

$$P(x_t | x_{t-1} x_{t-2} \dots x_1) = P(x_t | x_{t-1} x_{t-2} \dots x_{t-m})$$

$$\text{一阶: } P(x_t | x_{t-1} x_{t-2} \dots x_1) = P(x_t | x_{t-1})$$

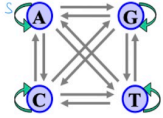
$$a_{kl} = P(x_t = s_l | x_{t-1} = s_k)$$



模型参数：状态转移矩阵。

Positive+	A	C	G	T
A	0.180	0.274	0.426	0.120
C	0.171	0.368	0.274	0.188
G	0.161	0.339	0.375	0.125
T	0.079	0.355	0.384	0.182

Negative-	A	C	G	T
A	0.300	0.205	0.285	0.210
C	0.322	0.298	0.078	0.302
G	0.248	0.246	0.298	0.208
T	0.177	0.239	0.292	0.292



$$\text{RATIO} = \frac{p(x | +\text{model})}{p(x | -\text{model})} = \frac{\prod_{i=0}^{L-1} p_+(x_{i+1} | x_i)}{\prod_{i=0}^{L-1} p_-(x_{i+1} | x_i)}$$

New sequence: **x=TGCAGCG**

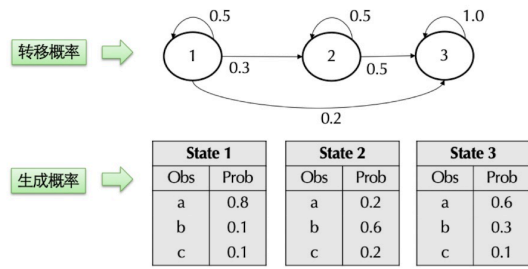
$$P(x | +\text{model}) = P_+(G|C) * P_+(C|G) * P_+(G|A) * P_+(A|C) * P_+(C|G) * P_+(G|T) \\ = 0.274 * 0.339 * 0.426 * 0.171 * 0.339 * 0.384 \\ = 0.000880819444$$

$$P(x | -\text{model}) = P_-(G|C) * P_-(C|G) * P_-(G|A) * P_-(A|C) * P_-(C|G) * P_-(G|T) \\ = 0.078 * 0.246 * 0.285 * 0.322 * 0.246 * 0.292 \\ = 0.00012648773$$

$$\text{RATIO} = P(x | +\text{model}) / P(x | -\text{model}) = 0.0008808 / 0.0001265 = 6.96 > 1$$

=> Sequence x more likely comes from CpG island regions

2. HMM



What's the probability of producing "a,a,b,c"?

$$P(a,a,b,c) \text{ via } 1,1,2,3 = 0.8 * 0.5 * 0.8 * 0.3 * 0.6 * 0.5 * 0.1 = 0.00288$$

$$P(a,a,b,c) \text{ via } 1,2,3,3 = 0.8 * 0.3 * 0.2 * 0.5 * 0.3 * 1.0 * 0.1 = 0.00072$$

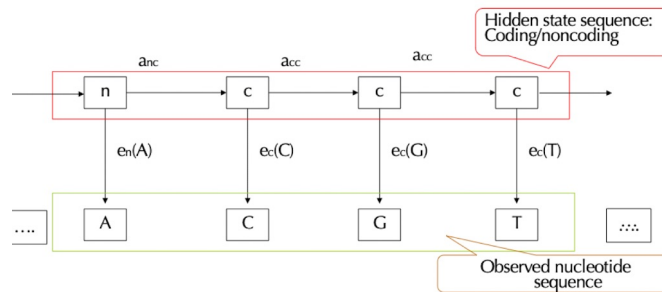
$$P(a,a,b,c) \text{ via } 1,3,3,3 = 0.8 * 0.2 * 0.6 * 1.0 * 0.3 * 1.0 * 0.1 = 0.00288$$

给定符号序列 $x = x_1 x_2 \dots x_L$

则通过状态序列 $\pi = \pi_1 \pi_2 \dots \pi_L$ 得到 x 的概率为

$$P(x, \pi) = a_{\pi_0 \pi_1} e_{\pi_1}(x_1) a_{\pi_1 \pi_2} e_{\pi_2}(x_2) \dots a_{\pi_{L-1} \pi_L} e_{\pi_L}(x_L) \\ = \prod_{i=0}^{L-1} a_{\pi_i \pi_{i+1}} e_{\pi_{i+1}}(x_{i+1})$$

π_0 为初始状态, e 为发射概率。



Decoding: 根据符号序列推测状态序列。

$$\pi^* = \underset{\pi}{\operatorname{argmax}} P(\pi | x)$$

$$P(\pi | x) = \frac{P(\pi, x)}{P(x)}$$

假设 $P(x)$ 与 π 无关

$$\text{则 } \pi^* = \underset{\pi}{\operatorname{argmax}} P(\pi, x)$$

$$= \underset{\pi}{\operatorname{argmax}} \prod_{i=0}^{L-1} a_{\pi_i \pi_{i+1}} e_{\pi_{i+1}}(x_{i+1})$$

dp: $F_k(\pi, x) : P(\pi_1 \dots \pi_k, x_1 \dots x_k)$ 最大值。

$$F_k(\pi, x) = \max_{\pi_{k-1}} \left\{ F_{k-1}(\pi, x) a_{\pi_{k-1} \pi_k} \right\} e_k(x_k)$$

	n	c
n	-0.097	-0.699
c	-0.398	-0.222

	A	C	G	T
n	-0.699	-0.523	-0.523	-0.699
c	-0.398	-0.699	-0.699	-0.699

	C	G	A	A	A	A
n	-0.62	-1.24	-2.036	-2.832	-3.628	-4.424
c	-1.40	-2.02	-2.337	-2.957	-3.577	-4.197
	A	A	A	T	C	G
n	-4.424	-5.22	-5.914	-6.534	-7.154	-7.774
c	-4.197	-4.817	-5.437	-6.058	-6.679	-7.299